Probabilistic Inventory models: Part 4

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A derivation of single-period Model based on the book "Design and Analysis of Lean Production Systems" authored Ronald G. Askin and Jeffrey B. Goldberg, John Wiley & Sons, 2002

- c=purchase (production) cost of each item
- p=price per item sold (related to revenue)
- s= salvage value of each unsold item
- D=demand, a random variable with probability density function $f(x)$
- L=shortage cost for each unmet item demand
- Q= number of units ordered (the decision variable)

Expected Profit

- $E(\text{profit})=(p-c)Q-L(D-Q)$, for $Q \leq D$
- $E(\text{profit})=pD-cQ+s(Q-D)$, for $Q>D$

$$E = p\int_0^Q xf(x)\mathrm{d}x + s\int_0^Q (Q-x)f(x)\mathrm{d}x + p\int_Q^\infty Qf(x)\,\mathrm{d}x - L\int_Q^\infty (x-Q)f(x)\mathrm{d}x - cQ$$

First term: revenue when demand is less than supply; Second term: salvage value
Third term: revenue when demand exceeds supply; Fourth term: shortage cost;
Fifth term: cost of goods

$$\frac{\mathrm{d}E}{\mathrm{d}Q}=\frac{\mathrm{d}}{\mathrm{d}Q}\left\{p\int_0^Q xf(x)\mathrm{d}x+s\int_0^Q\left(Q-x\right)f(x)\mathrm{d}x+p\int_Q^\infty Qf(x)\,\mathrm{d}x-L\int_Q^\infty\left(x-Q\right)f(x)\mathrm{d}x-cQ\right\}=0$$

Leibnitz Integral Rule

- Gottfried Wilhelm Leibnitz (1646–1716), a German polymath active as a mathematician, philosopher, scientist and diplomat
- Credited, alongside Sir Isaac Newton, with the creation of calculus in addition to many other branches of mathematics, such as binary arithmetic and statistics.
- Leibniz has been called the "last universal genius" due to his vast expertise across fields.

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{a(x)}^{b(x)} f(x,t)\mathrm{d}t\right) = f\{x,b(x)\}\frac{\mathrm{d}b(x)}{\mathrm{d}x} - f\{x,a(x)\}\frac{\mathrm{d}a(x)}{\mathrm{d}x} + \int_{a(x)}^{b(x)} \frac{\mathrm{d}}{\mathrm{d}x} f(x,t)\mathrm{d}t$$

Examples

Suppose

$$F(a)=\int_0^a \sin\left(x^2\right)\mathrm{d}x, \text{ then}$$

$$\frac{\mathrm{d}F}{\mathrm{d}a}=\sin\left(a^2\right)\frac{\mathrm{d}a}{\mathrm{d}a}=\sin\left(a^2\right)$$

Suppose

$$F(x)=\int_x^{2x} \ln\left(1+t^2\right)\mathrm{d}t, \text{ then}$$

$$\frac{\mathrm{d}F}{\mathrm{d}x}=\ln\left(1+4x^2\right)\frac{\mathrm{d}(2x)}{\mathrm{d}x}-\ln\left(1+x^2\right)\frac{\mathrm{d}(x)}{\mathrm{d}x}=\ln\left(1+4x^2\right)\frac{\mathrm{d}(2x)}{\mathrm{d}x}-\ln\left(1+x^2\right)$$

$$=2\ln\left(1+4x^2\right)-\ln\left(1+x^2\right)$$

More examples

Suppose

$$F(\theta)=\int_0^{\pi}\cos(\theta\sin t)\,\mathrm{d}t,\ \text{then}$$

$$\frac{\mathrm{d}F}{\mathrm{d}\theta}=\int_0^{\pi}\frac{\mathrm{d}}{\mathrm{d}\theta}\cos(\theta\sin t)\,\mathrm{d}t=\int_0^{\pi}-\sin t\sin(\theta\sin t)\,\mathrm{d}t$$

Suppose

$$F(x)=\int_x^{x^2} t\sin(tx)\,\mathrm{d}t,\ \text{then}$$

$$\frac{\mathrm{d}F}{\mathrm{d}x}=x^2\sin(x^3)\frac{\mathrm{d}x^2}{\mathrm{d}x}-x\sin(x^2)\frac{\mathrm{d}x}{\mathrm{d}x}+\int_x^{x^2} t^2\cos(tx)\,\mathrm{d}t$$

$$=2x^3\sin(x^3)-x\sin(x^2)+\int_x^{x^2} t^2\cos(tx)\,\mathrm{d}t$$

$$\frac{\mathrm{d}E}{\mathrm{d}Q} = \frac{\mathrm{d}}{\mathrm{d}Q}\left\{ p\int_0^Q xf(x)\mathrm{d}x + s\int_0^Q (Q-x)f(x)\mathrm{d}x + p\int_Q^\infty Qf(x)\,\mathrm{d}x - L\int_Q^\infty (x-Q)f(x)\mathrm{d}x - cQ \right\}$$

$$= pQf(Q) + s\int_0^Q f(x)\mathrm{d}x - pQf(Q) + p\int_Q^\infty f(x)\,\mathrm{d}x + L\int_Q^\infty f(x)\mathrm{d}x - c = 0$$

$$= sF(x) + (p+L)\{1-F(x)\} - c$$

Setting it equal to 0,

$$F(Q^*) = \frac{L+p-c}{L+p-s}$$

Numerator: Lost net revenue plus shortage penalty L, Denominator: Same plus loss from each unsold unit, *c*–*s*.

A simpler expression

- Shortage cost per unit=C_s =$L+p-c$
- Overstock cost per unit=C_o=$c-s$

$$F(Q^*) = \frac{C_s}{C_s + C_o}$$

Double derivative test

$$\frac{\mathrm{d}E}{\mathrm{d}Q} = \frac{\mathrm{d}}{\mathrm{d}Q}\left\{ p\int_0^Q xf(x)\mathrm{d}x + s\int_0^Q (Q-x)f(x)\mathrm{d}x + p\int_Q^\infty Qf(x)\,\mathrm{d}x - L\int_Q^\infty (x-Q)f(x)\mathrm{d}x - cQ \right\}$$

$$= pQf(Q) + s\int_0^Q f(x)\mathrm{d}x - pQf(Q) + p\int_Q^\infty f(x)\,\mathrm{d}x + L\int_Q^\infty f(x)\mathrm{d}x - c = 0$$

$$= sF(x) + (p+L)\{1-F(x)\} - c$$

$$\left.\frac{\mathrm{d}^2E}{\mathrm{d}Q^2}\right|_{Q=Q^*} = sf(Q^*) - (p+L)f(Q^*) = \text{negative}$$

No-Setup Model (Newsvendor Model)

Question: A newspaper operator needs to decide how many newspapers to stock each morning. Each newspaper is purchased at a cost of Rs. 12 and sold for Rs. 20. Maximum sales take place between 6:00 A.M and 7:00 A.M., making the demand effectively immediate. Any unsold newspapers at the end of the day are recycled, earning Rs. 2 per copy. Given that the daily demand follows a certain distribution, how many copies should be ordered each day?

(a) A normal distribution with mean 300 copies and standard deviation 20 copies.

(b) A discrete pdf, $f(D)$, defined as

D	200	250	300	320	340
$f(D)$	0.1	0.4	0.2	0.1	0.2

No-Setup Model (Newsvendor Model)

Solution: Each unsold copy will cost the owner Rs. 12–2 = Rs. 10

The penalty for running out of stock is Rs. 20–12= Rs. 8 per copy

Holding cost per unit, h = Rs 10 per copy per day

Penalty cost, p = Rs. 8 per copy per day

The critical ratio is determined as

$$\frac{p}{p+h} = \frac{8}{8+10} = 0.444 = \frac{C_s}{C_s + C_o}$$

No-Setup Model (Newsvendor Model)

Case (a): The demand *D* is *N*(300, 20). Using the standard normal table,

$$z = \frac{D - 300}{20}$$

From the normal tables

$$P\{z \leq -0.1\} = 0.444$$

or

$$\frac{y^* - 300}{20} = -0.1$$

$$\text{or } y^* = 298$$

Solving, y^* is obtained as 298.

No-Setup Model (Newsvendor Model)

Case (b): The demand D follows a discrete pdf, $f(D)$. First, the CDF $P\{D \leq y\}$ is determined as

y	200	250	300	320	340
$P\{D \leq y\}$	0.1	0.5	0.7	0.8	1.0

For the calculated critical ratio of 0.444,

$$P(D \leq 200) \leq 0.444 \leq P(D \leq 250)$$

Hence, y^* = 250 copies

Setup Model (s-S Policy)

Setup Model used in inventory management has following characteristics:

- A production or ordering process requires a setup before items can be produced or procured.
- Each setup incurs a fixed cost, regardless of the quantity produced or ordered.
- It applies when items are replenished multiple times over a planning horizon.

Commonly applied in production planning, make-to-stock systems, and inventory control in factories, warehouses and supply chains

Setup Model (s-S Policy)

As already mentioned, the present model differs from no-setup model in a way that a setup cost *K* is incurred. Using the same notation, the total expected cost per period is

$$E\{\bar{C}(y)\} = K + E\{C(y)\}$$

$$= K + h\int_0^y (y - D) f(D)\,\mathrm{d}D + p\int_y^\infty (D - y) f(D)\,\mathrm{d}D$$

The optimum value *y** must satisfy

$$P(y \le y^*) = \frac{p}{p + h}$$

Because *K* is constant, the minimum value of $E\{\bar{C}(y)\}$ must also occur at *y**

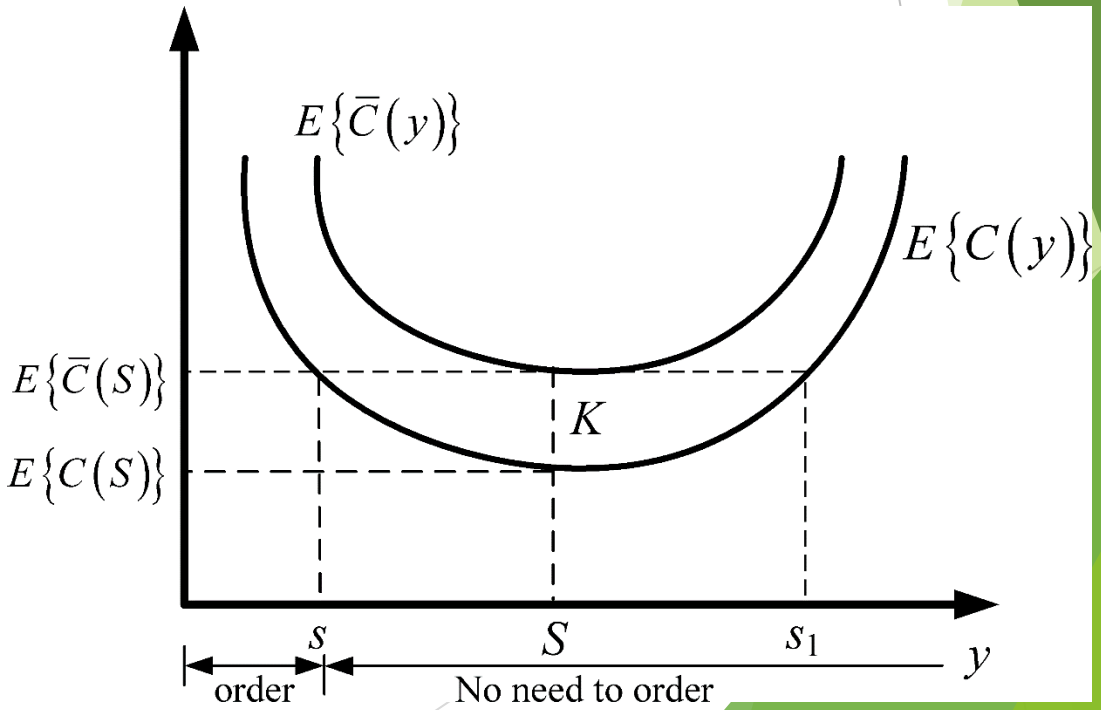
Setup Model (s-S Policy)

In Figure, $S = y^*$, and the value of s ($<S$) is determined from

$$E\{C(s)\} = E\{\bar{C}(S)\} = K + E\{C(S)\}, \quad s < S$$

The equation yields another value s_1 ($>S$), which is discarded. Assume that x is the amount on hand before an order is placed. How much should be ordered? This question is answered based on three conditions:

1. $x < s$
2. $s \le x \le S$
3. $x > S$

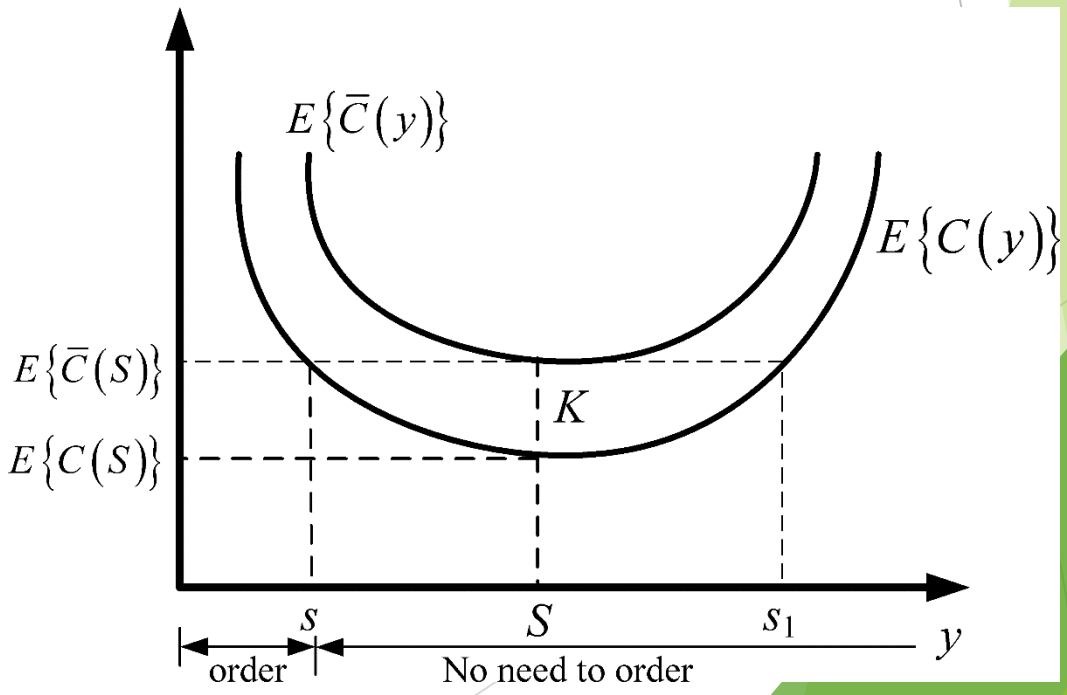


Setup Model (s-S Policy)

Case 1 (*r* < *s*): Because *x* is already on hand, its equivalent cost is given by $E\{C(x)\}$. If any additional amount $y - x (y > x)$ is ordered, the corresponding cost given *y* is $E\{\bar{C}(y)\}$, which includes the setup cost *K*. From Figure

$$\min_{y>x} E\{\bar{C}(y)\} = E\{\bar{C}(S)\} < E\{C(x)\}$$

Thus, the optimal inventory policy in this case is to order $S - x$ units.

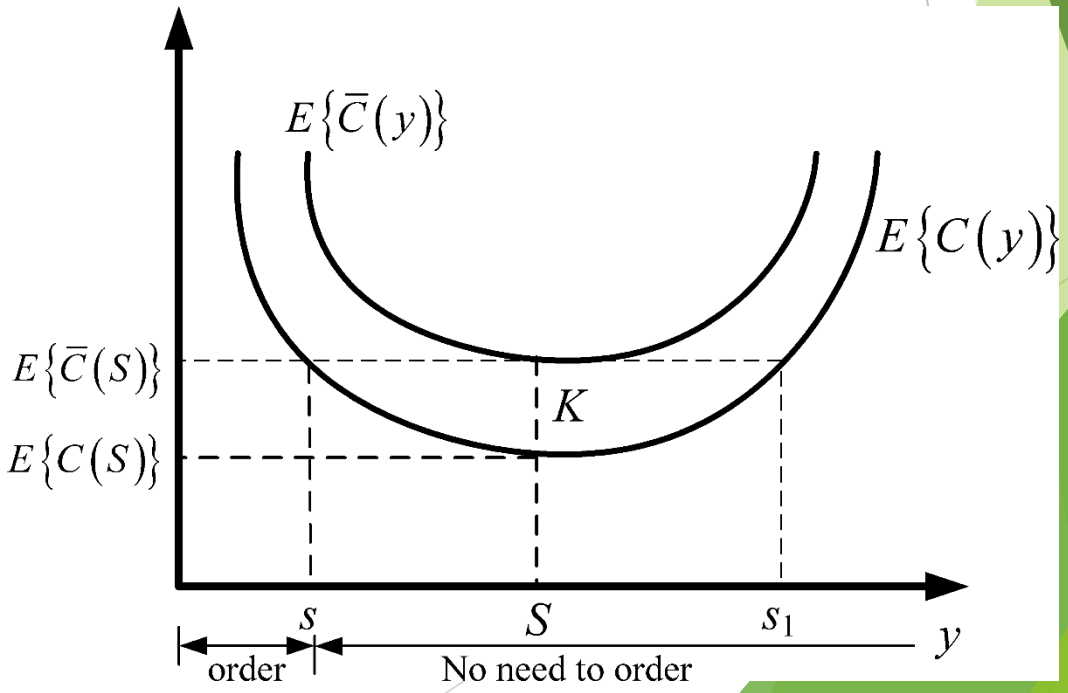


Setup Model (s-S Policy)

Case 2 ($s \le x \le S$): From Figure, we have

$$E\{C(x)\} \le \min_{y>x} E\{\bar{C}(y)\} = E\{\bar{C}(S)\}$$

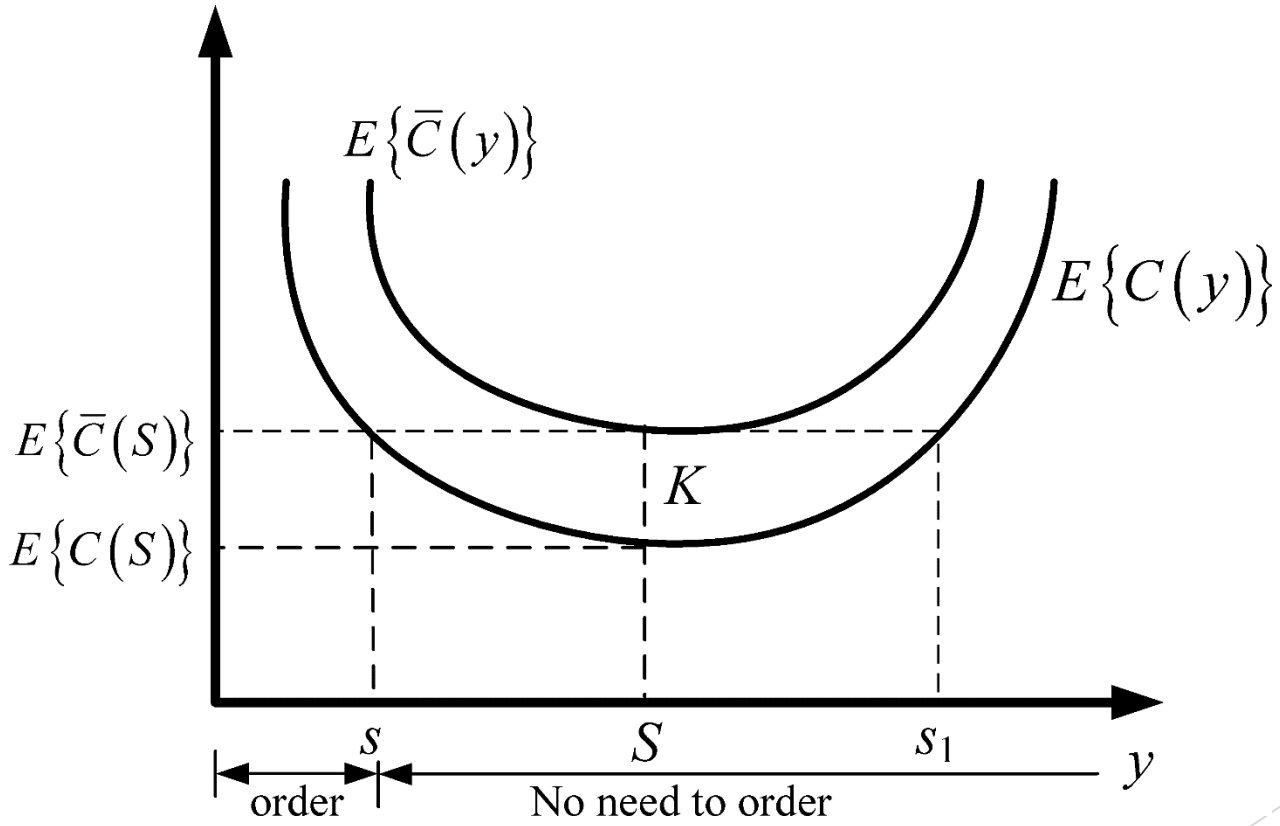
Thus, it is not advantageous to order in this case and $y^* = x$.



Setup Model (s-S Policy)

Case 3 ($x > S$): From Figure, for $y > x$

$$E\{C(x)\} < E\{\bar{C}(y)\}$$



Setup Model (s-S Policy)

This condition indicates that, as in case (2), it is not advantageous to place an order, i.e., $y^* = X$

The optimal inventory policy, frequently referred to as the *s*-*S* policy, is summarized as follows:

If $x < s$, order $S - X$

If $x \geq s$, do not order

The optimality of the *s*–*S* policy is guaranteed because the associated cost function is convex.

Setup Model (s-S Policy)

Example: An item experiences a one-time demand that occurs instantly at the beginning of a period. The demand follows a uniform probability distribution ranging from 0 to 10 units. The holding cost is Rs. 5 per unit for the duration of the period, and there is a penalty of Rs. 40 per unit for any unmet demand. Additionally, every order placed incurs a fixed cost of Rs. 100. Determine the optimal inventory policy for managing this item.

Solution: Given,

$p = 40, \quad h = 5$

The critical ratio y^* is obtained as

$$y^* = \frac{p}{p+h} = \frac{40}{40+5} = 0.89$$

Also,

$$P(D \le y^*) = \int_0^{y^*} \frac{1}{10}\,\mathrm{d}D = \frac{y^*}{10}$$

Thus,

$$S = y^* = \frac{0.89}{10} = 8.9$$

Expected cost function is

$$E\{C(y)\} = 5\int_0^{y} \frac{1}{10}(y - D)\,\mathrm{d}D + 45\int_y^{10} \frac{1}{10}(D - y)\,\mathrm{d}D$$

$$= \frac{5}{10}\left[yD - \frac{D^2}{2}\right]_0^{y} + \frac{45}{10}\left[\frac{D^2}{2} - yD\right]_y^{10}$$

$$= \left(\frac{5}{10} \times \frac{y^2}{2}\right) + \frac{45}{10}\left(\frac{10^2}{2} - \frac{y^2}{2} - 10y + y^2\right)$$

$$= 2.5y^2 - 45y + 225$$

The value of s is determined by solving

$$E\{C(s)\} = K + E\{C(S)\}$$

Hence,

$$2.5s^2 - 45s + 225 = 100 + 2.5S^2 - 45S + 225$$

For $S = 8.9$, the equation reduces to

$$2.5s^2 - 45s + 225 = 100 + 2.5S^2 - 45S + 225$$

$$\Rightarrow 2.5s^2 - 45s + 225 = 100 + 2.5 \times 8.9^2 - 45 \times 8.9 + 225$$

$$\Rightarrow s^2 - 18s + 41 = 0$$

Solving, s =15.32, 2.68

However, $s > S$ is discarded. Hence, s = 2.28 or 3 units.

Hence, if unit in hand is less than 3 units, then order the quantities to make 9 units in inventory.